Bounds for orthogonal fixed groups in odd characteristic

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This component supplies fixed-group bounds and two subspace counts for the orthogonal generation argument. Throughout q is odd and $\nu_2(H) = |\{h \in H : h^2 = 1\}|$, including the identity. We write $\mathrm{PCO}(V)$ for the full conformal orthogonal group modulo the scalar group over its ground field. This avoids confusion with the image of the isometry group, which can be a proper subgroup.

We retain the standard finite classical group orders, the faithful action of projective conformal groups on their derived simple groups, and Lawther–Liebeck–Seitz, Proposition 1.3 [1]. The latter gives $\mathrm{i}_2(\mathrm{Aut}\, S) < 2(Q^{N+r} + Q^{N+r-1})$ over $\mathbb{F}_Q$. The actual torus centralizers and projective eigenvalue degrees are acquired from the separately reviewed orthogonal torus component [3].

Lemma 1. *For $d \geq 2$ and a nondegenerate d-space over $\mathbb{F}_q$, put $e_d = \lfloor d^2/4 \rfloor$. Then*

$$\nu_2(\mathrm{PCO}_d(q)) \leq 3q^{e_d}, \qquad |\mathrm{PCO}_d(q)| < 3q^{d(d-1)/2}.$$

Proof. For $d \geq 5$, the derived projective group is simple, including the identifications in dimensions five and six. Its exponent $N + r$ is e_d. The faithful projective action and the retained bound give $\nu_2(\mathrm{PCO}_d(q)) < 2q^{e_d} + 2q^{e_d-1} + 1 \leq 3q^{e_d}$.

In dimension two the group is dihedral of order $2(q - \epsilon)$; its number of elements of square one is $q - \epsilon + 2 \leq q + 3$. In dimension three it is $\mathrm{PGL}_2(q)$, with q^2 nonidentity involutions. In dimension four the two models, already verified in the linear/unitary component [4], are

$$\mathrm{PGL}_2(q) \wr C_2, \qquad \mathrm{PGL}_2(q^2) \rtimes C_2.$$

The respective values of ν_2 are $q^4 + q^3 + 2q^2 - q + 1$ and $q^4 + q^3 + q + 1$, both at most $3q^4$. This proves the first assertion in every dimension needed here.

For odd d, the projective conformal group has the order of $\mathrm{SO}_d(q)$, less than $q^{d(d-1)/2}$. For even $d = 2s$, it has the order of the full isometry group,

$$2q^{s(s-1)}(q^s - \epsilon)\prod_{j=1}^{s-1}(q^{2j} - 1) < 2(1 + q^{-s})q^{d(d-1)/2} \leq \tfrac{8}{3}q^{d(d-1)/2}.$$

The ordinary order formulas also include dimension two. □

Proposition 2. *Let V have dimension $2m$, $m \geq 3$. Suppose a conformal map a has square cI, with multiplier c, and its two eigenspaces over the algebraic closure both have dimension m. Then*

$$\nu_2(C_{\mathrm{PCO}(V)}([a])) \leq 24q^{2\lfloor m^2/4 \rfloor}.$$

If c is a square and the two eigenspaces instead have unequal dimensions $r, s \geq 2$, then

$$\nu_2(C_{\mathrm{PCO}(V)}([a])) \leq 18q^{e_r+e_s}.$$

Proof. First suppose c is a square. Rescale a to have square one. Its two eigenspaces are nondegenerate and orthogonal. A map centralizing $[a]$ sends a to a or $-a$, since it preserves the multiplier. It therefore preserves the two eigenspaces, or interchanges them. Projecting its restrictions to their projective conformal groups has kernel of order at most two: a scalar on each block must have the same square, because the two multipliers agree, and a common scalar is already factored out.

When an interchange is possible the two forms are similar. Writing H for either projective conformal group, the image is a subgroup of $H \wr C_2$. There are $\nu_2(H)^2$ elements of square one in the base, and exactly $|H|$ in the interchange coset, since its two coordinates must be mutual inverses. A group extension with kernel of order at most two has at most twice as many elements of square one as its image. Lemma 1 thus gives

$$\nu_2(C) \le 2\big(9q^{2e_m} + 3q^{m(m-1)/2}\big) \le 24q^{2e_m}.$$

If the forms are not similar, omit the interchange coset and use the product bound. Unequal dimensions also forbid interchange, giving $2(3q^{e_r})(3q^{e_s})$ as stated.

Now suppose c is a nonsquare. The algebra $K = \mathbb{F}_q[a]$ is $\mathbb{F}_{q^2}$. The map a is self-adjoint for the underlying bilinear form, because $a^* = ca^{-1} = a$. The nondegenerate trace pairing then writes that form uniquely as the trace of a nondegenerate symmetric K-bilinear form on the m-dimensional K-space V. A projective centralizer acts either linearly or by the nontrivial field automorphism of K; the induced projective group embeds in the automorphism group of the corresponding derived orthogonal group over K. Its kernel has order at most two. Indeed a scalar $z \in K^\times$ induces an allowed similitude over $\mathbb{F}_q$ only if $z^2 \in \mathbb{F}_q^\times$, and there are exactly two such cosets modulo $\mathbb{F}_q^\times$.

For $m \ge 5$, simplicity and the retained involution bound give $\nu_2(C) \le 6q^{2e_m}$. The same argument applies when $m = 3$, using $\mathrm{P}\,\Omega_3(K) \simeq \mathrm{PSL}_2(K)$. For $m = 4$ of minus type it uses $\mathrm{P}\,\Omega_4^-(K) \simeq \mathrm{PSL}_2(q^4)$, and gives $6q^8$.

The plus case in dimension four needs its two simple factors. Put $Q = q^2$ and $A = \mathrm{PSL}_2(Q)$; the derived projective group is $A \times A$. Its full automorphism group is $\mathrm{Aut}(A) \wr C_2$. We have $\nu_2(\mathrm{Aut}\,A) \le 3Q^2$ and $|\,\mathrm{Aut}\,A| = f_Q Q(Q^2-1) < Q^4$, where $Q = p^{f_Q}$ and $f_Q \le Q$. Thus

$$\nu_2(C) \le 2\big(9Q^4 + Q^4\big) = 20q^8.$$

These bounds all imply the stated constant 24. □

Proposition 3. *For a natural involutory reversing automorphism of any of the chosen orthogonal elements over odd $\mathbb{F}_q$, its fixed subgroup in the simple group has at most*

$$\begin{cases} 24q^{2\lfloor m^2/4 \rfloor}, & \dim V = 2m, \\ 18q^{m(m+1)/2}, & \dim V = 2m+1 \end{cases}$$

elements of square one. This assertion does not count triality reversers which lack a representative on the natural module.

Proof. Let X be the natural representative of the chosen torus element. Projective inversion initially gives $aXa^{-1} = \lambda X^{-1}$, with $\lambda^2 = 1$. On the large block, the acquired degree lemma forces the field exponent of a to be k, where the block has dimension $2k$. Since the eigenvalue α satisfies $\alpha^{q^k} = \alpha^{-1}$ exactly, the scalar is $\lambda = 1$. Consequently the restriction is $z\,\mathrm{Frob}^k$. Its square and conformal multiplier are both $\mathrm{N}_{\mathbb{F}_{q^{2k}}/\mathbb{F}_{q^k}}(z)$, which lies in $\mathbb{F}_q^\times$. The same calculation on the negative two-plane gives the same scalar; on a fixed line its multiplier is the square of its line scalar. Thus $a^2 = cI$, with multiplier c.

If c is a square, on every negative $2k$-block a scalar multiple of a is a semilinear involution for the quadratic extension over $\mathbb{F}_{q^k}$. Its fixed and negative spaces both have dimension k.

This follows also by solving $zv^{q^k} = \pm v$: norm one is precisely the solvability condition, and each solution space is one-dimensional over $\mathbb{F}_{q^k}$. The dimensions are therefore m, m in even dimension and $m, m+1$ in odd dimension. In the latter case the line forces c to be a square. Apply Proposition 2; taking a subgroup cannot increase its number of elements of square one. □

Proposition 4. *For odd q the following strict lower bounds hold:*

$$\mathrm{i}_2(P\Omega_8^+(q)) > q^{16}/4, \qquad \mathrm{i}_2(P\Omega_8^-(q)) > q^{16}/2.$$

Proof. In either space let W run over the nondegenerate four-dimensional subspaces of plus type. Put $t_W = -I$ on W and $t_W = I$ on its orthogonal complement. Its determinant is one and its spinor norm is the discriminant of the four-dimensional plus form, a square. Thus $t_W \in \Omega(V)$. We use here the standard spinor norm of $-I$ on a nondegenerate subspace, as in [2, Definition 3.1]. Witt's extension theorem makes the full isometry group transitive on these subspaces, with stabilizer $O(W) \times O(W^\perp)$.

In plus dimension eight the complement also has plus type. The projective images of t_W and $t_{W'}$ agree exactly when $W' = W$ or $W' = W^\perp$: the scalar identifying two isometries must be 1 or -1. No nondegenerate W equals its complement. Hence the resulting distinct involutions number

$$\frac{|O_8^+(q)|}{2|O_4^+(q)|^2} = \frac{q^8(q^2+1)^2(q^4+q^2+1)}{4} > \frac{q^{16}}{4}.$$

In minus dimension eight the complement has minus type, while $-I$ has nonsquare spinor norm and does not lie in $\Omega(V)$. There is no projective identification between the displayed involutions. Their number is

$$\frac{|O_8^-(q)|}{|O_4^+(q)|\,|O_4^-(q)|} = \frac{q^8(q^4+1)(q^4+q^2+1)}{2} > \frac{q^{16}}{2}.$$

The count is of actual subspaces, so it does not require transitivity of the smaller omega group on this set. □

Scope. The constants deliberately include the identity, possible projective order-four lifts, a nonsplit quadratic-field form, and interchange cosets. No central quotient is estimated by counting only involutions in its covering group. The separate covering and total-involution estimates remain necessary for a generation conclusion.

References

[1] R. Lawther, M. W. Liebeck and G. M. Seitz, *Fixed point ratios in actions of finite exceptional groups of Lie type*, Pacific Journal of Mathematics 205 (2002), Proposition 1.3. The full automorphism bound, including the identity adjustment here, is retained, not reproved.

[2] S. H. Murray and C. M. Roney-Dougal, *The spinor norm and homomorphism algorithms for classical groups*, preprint dated 2 May 2008, Definition 3.1 and Lemma 3.2.

[3] Human Compression project, *Orthogonal tori and their projective orders*, accepted component with independent review.

[4] Human Compression project, *Linear and unitary groups*, accepted component; the precise dimension-four projective conformal models and their counts are acquired.